

Scientific Skills Exercise

Analyzing Polypeptide Sequence Data

Are Rhesus Monkeys or Gibbons More Closely Related to Humans? In this exercise, you will look at amino acid sequence data for the β polypeptide chain of hemoglobin, often called β -globin. You will then interpret the data to hypothesize whether the monkey or the gibbon is more closely related to humans.

How Such Experiments Are Done Researchers can isolate the polypeptide of interest from an organism and then determine the amino acid sequence. More frequently, the DNA of the relevant gene is sequenced, and the amino acid sequence of the polypeptide is deduced from the DNA sequence of its gene.

Data from the Experiments In the data below, the letters give the sequence of the 146 amino acids in β -globin from humans,

Species	Alignment of Amino Acid Sequences of β -globin					
Human	1	VHLTPEEKSA	VTALWGKVVN	DEVGGEALGR	LLVYPWTQR	FFESFGDLST
Monkey	1	VHLTPEEKNA	VTTLWGKVVN	DEVGGEALGR	LLLVYPWTQR	FFESFGDLSS
Gibbon	1	VHLTPEEKSA	VTALWGKVVN	DEVGGEALGR	LLVYPWTQR	FFESFGDLST
Human	51	PDAVMGNPKV	KAHGKKVLGA	FSDGLAHLDN	LKGTFA TLSE	LHCDKLHVDP
Monkey	51	PDAVMGNPKV	KAHGKKVLGA	FSDGLNHLDN	LKGTFAQLSE	LHCDKLHVDP
Gibbon	51	PDAVMGNPKV	KAHGKKVLGA	FSDGLAHLDN	LKGTFAQLSE	LHCDKLHVDP
Human	101	ENFRL LGNVL	VCVLAHHFGK	EFTPPVQAAY	QKVVAGVANA	LAHKYH
Monkey	101	ENFRL LGNVL	VCVLAHHFGK	EFTPPVQAAY	QKVVAGVANA	LAHKYH
Gibbon	101	ENFRL LGNVL	VCVLAHHFGK	EFTPPVQAAY	QKVVAGVANA	LAHKYH

Data from Human: <http://www.ncbi.nlm.nih.gov/protein/AAA21113.1>; rhesus monkey: <http://www.ncbi.nlm.nih.gov/protein/122634>; gibbon: <http://www.ncbi.nlm.nih.gov/protein/122616>

► Human



► Rhesus monkey



► Gibbon



rhesus monkeys, and gibbons. Because a complete sequence would not fit on one line here, the sequences are divided into three segments: amino acids 1–50, 51–100, and 101–146. The sequences for the three different species are aligned so that you can compare them easily. For example, you can see that for all three species, the first amino acid is V (valine) and the 146th amino acid is H (histidine).

INTERPRET THE DATA

- Scan the monkey and gibbon sequences, letter by letter, circling any amino acids that do not match the human sequence. (a) How many amino acids differ between the monkey and the human sequences? (b) Between the gibbon and human?
- For each nonhuman species, what percent of its amino acids are identical to the human sequence of β -globin?
- Based on these data alone, state a hypothesis for which of these two species is more closely related to humans. What is your reasoning?
- What other evidence could you use to support your hypothesis?

➔ **Instructors:** A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

PROBLEM-SOLVING EXERCISE

Are you a victim of fish fraud?



When buying salmon, perhaps you prefer the more expensive wild-caught Pacific salmon (*Oncorhynchus* species) over farmed Atlantic salmon (*Salmo salar*). But studies reveal that about 40% of the time, you aren't getting the fish you paid for!

In this exercise, you will investigate whether a piece of salmon has been fraudulently labeled.

Your Approach The principle guiding your investigation is that DNA sequences from within a species or from closely related species are more similar to each other than are sequences from more distantly related species.

Your Data You've been sold a piece of salmon labeled as coho salmon (*Oncorhynchus kisutch*). To see whether your fish was labeled correctly, you will compare a short DNA sequence from your sample with known sequences from the same gene for three salmon species. The sequences are as follows:

Sample labeled as <i>O. kisutch</i> (coho salmon)	5' -CGGCACCGCCCTAAGTCTCT-3'
<i>O. kisutch</i> (coho salmon)	5' -AGGCACCGCCCTAAGTCTAC-3'
<i>O. keta</i> (chum salmon)	5' -AGGCACCGCCCTGAGCCTAC-3'
<i>Salmo salar</i> (Atlantic salmon)	5' -CGGCACCGCCCTAAGTCTCT-3'

Data from the International Barcode of Life.

Your Analysis

- Circle any bases in the known sequences that do not match the sequence from your fish sample.
- How many bases differ between (a) *O. kisutch* and your fish sample? (b) *O. keta* and the sample? (c) *S. salar* and the sample?
- For each known sequence, what percentage of its bases are identical to your sample?
- Based on these data alone, state a hypothesis for the species identity of your sample. What is your reasoning?

➔ **Instructors:** A version of this Problem-Solving Exercise can be assigned in Chapter 5 of **Mastering Biology**. A more extensive "Solve It" tutorial is in Chapter 26 of **Mastering Biology**.